



Echinococcus granulosus and Hymenolepis nana

Microbiology GIT Block · MEDucation by Mattin Majid

BEFORE YOU READ — ANSWER FROM MEMORY

1. What are the definitive and intermediate hosts of *Echinococcus granulosus*, and why does the cycle "dead-end" in humans?
2. What is hydatid sand, and why is cyst rupture dangerous?
3. What makes *Hymenolepis nana* unique among the cestodes regarding its life cycle and host requirement?

Guessing wrong before reading still improves retention — attempt all three, then check as you go.

1 *Echinococcus granulosus* — the dog tapeworm

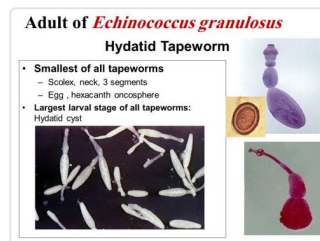
- Common name: **dog tapeworm**; causes **echinococcosis (hydatid disease)**; a **cestode**
- History: hydatid cysts described by Hippocrates; adult worm described by Hartmann (1695) in dog small intestine; larval (hydatid cyst) form recognized by Goeze (1782)
- **Distribution**: worldwide, most extensive in **sheep/cattle-raising areas** (Australia, Africa, South America); also common in Europe, China, Middle East; more in **temperate** than tropical regions

Host	Role
Definitive hosts	Canids, felids, hyaenids (dogs, wolves, foxes) — adult worm lives in jejunum/duodenum , lives 6-30 months
Intermediate hosts	Herbivores (sheep, goat, cattle, horse) and humans — harbor the larval stage (hydatid cyst)

Humans are an accidental intermediate host: ingestion of eggs → larval (hydatid cyst) stage. Since human hydatid cysts are unlikely to be eaten by dogs, **the cycle is a dead end in humans**.



Adult *E. granulosus* (small white worms) in a dog's small intestine.



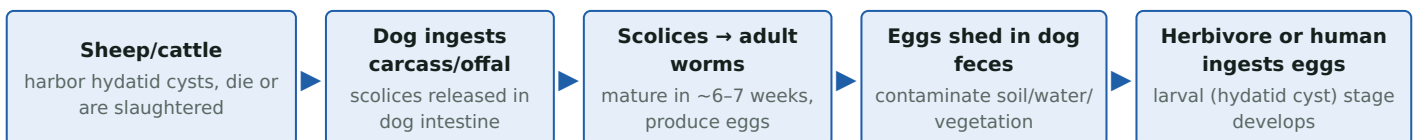
Labelled adult morphology — scolex, neck, 3 segments; egg; hydatid cyst (largest larval stage of any tapeworm).



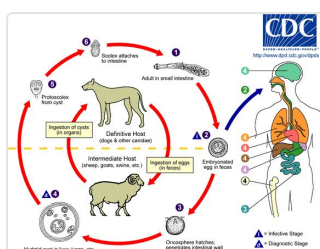
Zoonotic transmission: definitive hosts shed eggs; humans and herbivores ingest them.

2 Transmission and life cycle

- Humans ingest tapeworm eggs shed by definitive hosts — via **contaminated fruits, vegetables, herbs, water**
- In endemic areas, dogs often have ready access to **viscera from slaughtered livestock** (sheep slaughtering close to free-ranging dogs) — this sustains the cycle



Two-host life cycle: definitive host = carnivores; intermediate hosts = herbivores and (accidentally) humans.



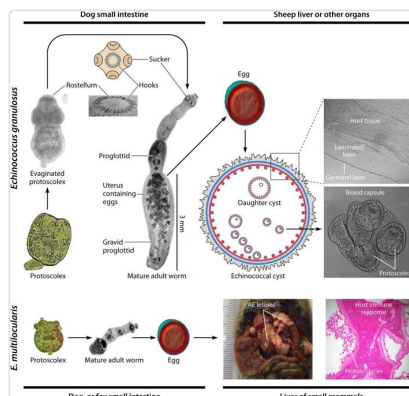
Two-host life cycle (CDC-style).



Dogs scavenging livestock carcasses sustains transmission.

3 Pathogenesis — evolution of the hydatid cyst

- At the deposition site, the embryo slowly develops into a **hollow, fluid-filled bladder** — the **hydatid cyst** (Greek *hydatis* = drop of water)
- Contains **thousands of individual scolices** plus **daughter cysts**; scolices lying at the bottom are called "**hydatid sand**"
- Cyst reaches **0.5-1 cm diameter in ~6 months**; host reaction forms a **fibrous capsule** around it
- Cyst fluid contains **parasite antigens** that sensitize the host — **rupture (spontaneous, trauma, or surgical)** can cause **life-threatening anaphylactic shock** and **widespread protoscolex seeding**

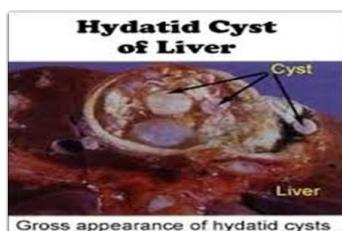


Hydatid cyst structure: germinal layer, laminated membrane, brood capsules, protoscolices, and daughter cysts.

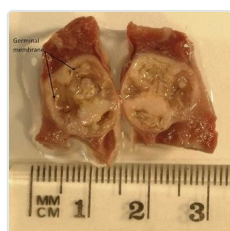
4 Clinical manifestations — a slow-growing "tumor"

Site	Frequency / Feature
Liver	~ 60-75% — first filter for the embryo; can cause obstructive jaundice
Lung	~ 20-25% — second filter (embryos that pass the liver reach the lung via right heart/pulmonary capillaries); bronchial cysts may produce a pulmonary abscess
Other (spleen, kidney, eye, brain, bone)	A few embryos enter systemic circulation and lodge elsewhere; brain cysts → raised ICP/epilepsy; kidney cysts → renal dysfunction; bone cysts → pathological collapse; eye → blindness

- Incubation:** months to years (20-30 years documented for slow-growing cysts)
- Most infections are asymptomatic**, found incidentally; disease appears once the cyst is **large enough to cause pressure/obstructive effects**
- Rupture** of a previously silent cyst can trigger **sudden anaphylactic shock**



Hydatid cyst of liver — gross appearance.



Bisected hydatid cysts with daughter cysts (scale in cm).



Ocular hydatid cyst.

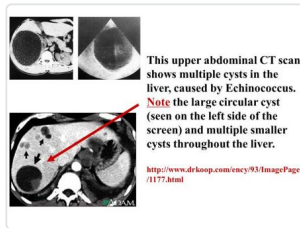


Cerebral hydatidosis.

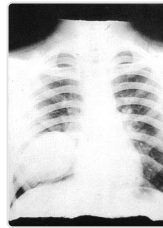
5 Diagnosis and treatment

	Detail
Imaging	USG, CT, MRI — diagnostic in most cases of cystic echinococcosis
Serology	Antigen detection by counter immunoelectrophoresis (CIEP); blood eosinophilia 20-25%
Protoscolex detection	In urine/sputum, via lactophenol cotton blue (LPCB) staining; aspirated cyst fluid shows protoscolices (~100 µm) with typical hooklets
Molecular	PCR
Treatment — early	PAIR: Puncture, Aspiration, Injection, Reaspiration

Treatment — complicated cysts	Surgery is treatment of choice; must avoid rupture (germinal-layer fragments can seed new cysts); recurrence after surgery is common
Chemotherapy	Albendazole , pre- and post-operatively, after curative surgery



CT: hydatid cysts in the liver.



Chest X-ray: pulmonary hydatid cyst.

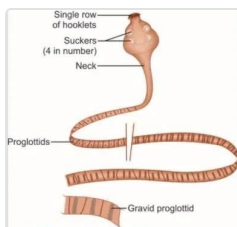


Prevention: hand hygiene, safe carcass disposal, avoiding dog access to offal.

6 *Hymenolepis nana* — the dwarf tapeworm

- Common name: **dwarf tapeworm**; name from Greek *hymen* (membrane) + *lepis* (rind) describing the egg membrane, and *nanus* (dwarf) for its small size
- First discovered by **Bilharz (1857)**; **cosmopolitan**, more common in **warm climates**; most common in **school children and institutional populations**
- Smallest and most common tapeworm of the human intestine**
- Unique**: the **only cestode that completes its entire life cycle in one host — humans** (no intermediate host needed)

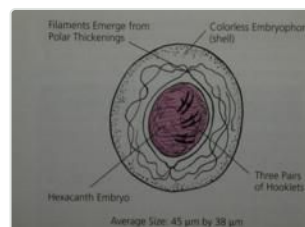
Feature	Detail
Adult worm	Lives in the ileum ; 5-45 mm long , <1 mm thick ; scolex has 4 suckers + retractile rostellum with a single row of hooklets ; strobila of 200+ proglottids (broader than long)
Egg	Roughly spherical/ovoid, 30-40 µm ; thin colorless outer membrane + inner embryophore enclosing the hexacanth oncosphere ; space between membranes has yolk granules and 4-8 polar filaments ; immediately infective , survives <10 days outside the host
Host	Man only — no intermediate host



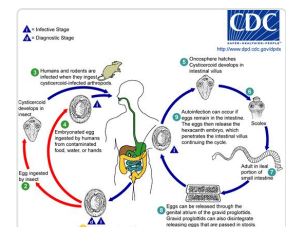
Labelled adult morphology: scolex with hooklets, neck, gravid proglottids.



Egg on microscopy.



Labelled egg: polar filaments emerging from polar thickenings, hexacanth embryo.



	Detail
Clinical features	Usually asymptomatic ; heavy infection → nausea, anorexia, abdominal pain, diarrhea, irritability; pruritus (allergic response) may occur; more common in children
Diagnosis	Characteristic eggs in feces on direct microscopy; ELISA
Treatment	Praziquantel — drug of choice, acts on both adult worms and cysticercoids in the intestinal villi
Prophylaxis	Personal hygiene, sanitation, avoid contaminated food/water, rodent control

P Must-remember pearls

- **E. granulosus**: dog = definitive host, sheep/cattle/human = intermediate; human cycle is a **dead end**; larva = **unilocular hydatid cyst**.
- **Hydatid sand** = free protoscolices at the cyst base; **rupture** → **anaphylaxis** + seeding of new cysts — the single most dangerous complication.
- Cyst distribution: **liver (60-75%, 1st filter) > lung (20-25%, 2nd filter)** > other sites (spleen, kidney, brain, bone, eye).
- Early disease → **PAIR**; complicated cysts → **surgery** (avoid rupture) + **perioperative albendazole**; recurrence is common.
- **H. nana** is the **only cestode completing its full life cycle in one host (man)** — no intermediate host, eggs are **immediately infective**; **praziquantel** treats both adults and cysticercoids.

Q Test yourself

1. A sheep farmer develops a slow-growing liver cyst discovered incidentally on ultrasound. He keeps dogs that have access to offal from home slaughter. What is the most likely diagnosis and why does this cycle dead-end in him?

- | | |
|---|--|
| A. Echinococcus granulosus — humans are an accidental intermediate host and hydatid cysts are not eaten by dogs | B. Hymenolepis nana — humans are the only host required |
| C. Ascaris lumbricoides — humans are the definitive host | D. Taenia solium — pigs are required as intermediate hosts |

2. A patient undergoes surgery for a hydatid cyst and the surgeon takes great care not to rupture it. Why?

- | | |
|---|--|
| A. Rupture can cause anaphylactic shock and seed new cysts from spilled protoscolices | B. Rupture causes immediate bacterial sepsis |
| C. Rupture releases a neurotoxin causing seizures | D. Rupture is painless and has no clinical consequence |

3. Which organ is most commonly affected by hydatid cysts, and what anatomic principle explains this distribution?

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|--|--|
| A. Liver — it is the first capillary filter the embryo encounters via the portal circulation | B. Lung — it is the first filter encountered |
| C. Brain — embryos preferentially cross the blood-brain barrier | D. Kidney — embryos are filtered renally first |

4. What is the unique feature of the Hymenolepis nana life cycle compared to other cestodes discussed?

- | | |
|--|--|
| A. It completes its entire life cycle in a single host (humans), with no intermediate host | B. It requires both a snail and a fish intermediate host |
| C. It requires ingestion of undercooked beef | D. It is transmitted solely by mosquito bite |

5. Which drug is the treatment of choice for Hymenolepis nana, and why is it preferred?

- | | |
|--|---|
| A. Praziquantel — effective against both adult worms and cysticercoids in the intestinal villi | B. Albendazole — effective only against adult worms |
| C. Metronidazole — effective against protozoa, not cestodes | D. Ivermectin — effective against nematodes, not cestodes |

ANSWERS — COVER UNTIL YOU'VE COMMITTED

1 — A — Dogs are the definitive host; humans are an accidental intermediate host, and since human hydatid cysts are not eaten by dogs, transmission cannot continue.

2 — A — Cyst fluid contains antigens that can trigger anaphylaxis, and spilled protoscolices/germinal layer fragments can seed secondary cysts.

3 — A — The liver acts as the first filter for the embryo via the portal circulation, accounting for 60-75% of cysts; the lung is the second filter.

4 — A — H. nana is unique among cestodes for completing its full life cycle in one host, humans, without needing an intermediate host.

5 — A — Praziquantel is preferred because it acts on both the adult worms and the cysticercoid larvae within the intestinal villi.

SPACED REVIEW

☐ Day 1

☐ Day 3

☐ Day 7

☐ Day 21

☐ Pre-exam